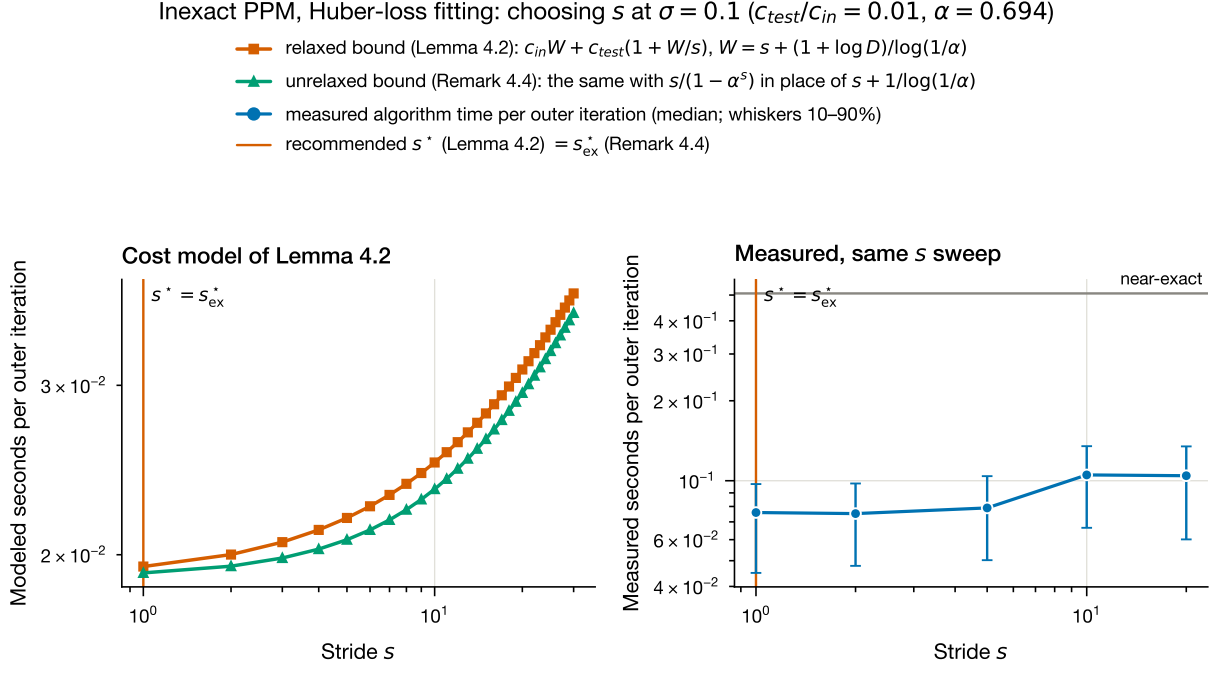
Inexact PPM, Huber-loss fitting: choosing s at $\sigma = 0.1$ ($c_{test}/c_{in} = 0.01$, $\alpha = 0.694$)
relaxed bound (Lemma 4.2): $c_{in}W + c_{test}(1 + W/s)$, $W = s + (1 + \log D)/\log(1/\alpha)$
unrelaxed bound (Remark 4.4): the same with $s/(1 - \alpha^s)$ in place of $s + 1/\log(1/\alpha)$
measured algorithm time per outer iteration (median; whiskers 10–90%)
recommended $s^\star$ (Lemma 4.2) $= s^\star_{\text{ex}}$ (Remark 4.4)
Cost model of Lemma 4.2
$s^\star = s^\star_{\text{ex}}$
Modeled seconds per outer iteration
3×10^{-2}
2×10^{-2}
10^0
10^1
Stride s
Measured, same s sweep
near-exact
$s^\star = s^\star_{\text{ex}}$
Measured seconds per outer iteration
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
6×10^{-2}
4×10^{-2}
10^0
10^1
Stride s